Topics Covered

Electronics Fundamentals

- Voltage, Current, Resistance
- Ohm's Law
- Kirchhoff's Laws
- Circuit Analysis
- Resistors, Capacitors, and Inductors
- Diodes and Transistors
- Operational Amplifiers
- Analog and Digital Electronics

Digital Systems

- Number Systems
- Logic Gates
- Boolean Algebra
- Combinational Circuits
- Sequential Circuits
- Flip-Flops and Registers

Embedded Systems

- Introduction to Embedded Systems
- Embedded Hardware Architecture
- Microcontrollers and Microprocessors
- GPIO Programming
- Timers and Interrupts
- ADC and DAC
- Communication Protocols

    - UART
    - SPI
    - I2C
    - CAN
- Real-Time Systems Basics

Learning Path

1. Start with Basic Electronics.
2. Study Digital Logic Design.
3. Learn Microcontroller Fundamentals.
4. Understand Embedded Hardware Architecture.
5. Practice Embedded Programming Concepts.
6. Explore Communication Protocols and Applications.

Recommended Tools

- Proteus
- KiCad
- STM32CubeIDE
- Arduino IDE
- MPLAB X
- VS Code